Breast Cancer Diagnosis Classification Report

Wisconsin Diagnostic Breast Cancer Data

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Abstract

This report analyzes the Wisconsin Diagnostic Breast Cancer data as a binary classification problem. The goal is to predict whether a tumor is malignant or benign using 30 numeric measurements computed from digitized images of breast-mass samples. The analysis combines exploratory data analysis, correlation diagnostics, principal component analysis, logistic regression, linear discriminant analysis, lasso logistic regression, and elastic net logistic regression. The dataset contains 569 observations, including 357 benign cases and 212 malignant cases. Across the analysis, tumor size and shape-irregularity measurements provide the strongest separation between the two diagnosis groups. On the holdout test set, elastic net logistic regression gives the best overall result, with accuracy 0.971, sensitivity 0.937, specificity 0.991, and AUC 0.992. Repeated cross-validation supports the same conclusion, with mean accuracy 0.972 and mean AUC 0.993 for elastic net. The results show that the WDBC measurements contain strong classification signal, although the fitted model should be viewed as a statistical screening tool rather than a clinical decision rule.

1. Data and Objective

The main objective is to classify tumors as malignant or benign using numeric cell-nucleus measurements. This is a supervised binary classification problem. Malignant cases are coded as $Y = 1$ and benign cases as $Y = 0$, so the fitted models estimate the probability $P(Y = 1 \mid X)$, where X represents the predictor measurements.

The data file contains an ID variable, a diagnosis label, and 30 numeric predictors. The predictors are based on ten feature families: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. For each feature family, the data include the mean value, the standard error, and the worst or largest value for the case. The raw file does not include column headers, so the predictor names used in the analysis were assigned from the standard WDBC feature structure, such as mean radius, radius standard error, and worst radius. The ID column was removed because it is an observation identifier rather than a predictive feature.

Table 1. Class balance in the WDBC data.

Diagnosis	Count	Proportion
Benign	357	0.627
Malignant	212	0.373

Table 2. Main feature groups used in the analysis.

Variable	Meaning
diagnosis	Original class label: benign or malignant
malignant	Binary response used for modeling: 1 = malignant, 0 = benign
radius	Mean distance from center to points on the perimeter
texture	Standard deviation of gray-scale values
perimeter	Size of the cell-nucleus boundary
area	Area of the cell nucleus
smoothness	Local variation in radius lengths
compactness	Perimeter squared divided by area minus one
concavity	Severity of concave portions of the contour
concave_points	Number/severity of concave portions of the contour
symmetry	Symmetry of the cell nucleus
fractal_dimension	Coastline approximation of the nucleus boundary

2. Data Preparation

The diagnosis label was stored as a factor for classification tasks and as a 0/1 response for logistic regression. The data did not contain missing values, so no observations were removed and no imputation was needed. All predictors used for modeling were numeric. The only categorical field in the modeling data is the diagnosis label.

Standardization is important in this project because the predictors are measured on very different scales. For example, area measurements can be hundreds or thousands of units, while smoothness and fractal-dimension measurements are small decimals. PCA, LDA, lasso logistic regression, and elastic net logistic regression were fit using standardized predictors, so that the analysis was not driven simply by the largest-scale variables. The smaller logistic baseline was kept on the original predictor scale to make its odds ratios easier to interpret.

Table 3. Training and test split summary.

Dataset	Observations	Malignant	Benign
Training	399	149	250
Test	170	63	107

3. Exploratory Analysis

The first part of the analysis compares selected predictor summaries across benign and malignant cases. Malignant tumors generally have larger values on size-related and contour-irregularity measurements. For example, mean radius is 12.147 for benign cases and 17.463 for malignant cases. Mean area is 462.790 for benign cases and 978.376 for malignant cases. The difference is even larger for worst area, where the benign mean is 558.899 and the malignant mean is 1422.286. Concavity and concave-points measurements also show a clear shift upward for malignant cases.

Table 4. Selected descriptive statistics by diagnosis group.

Variable	Benign	Malignant	Malignant - Benign
Mean radius	12.147	17.463	5.316
Mean texture	17.915	21.605	3.690
Mean area	462.79	978.38	515.59
Mean smoothness	0.092	0.103	0.010
Mean concavity	0.046	0.161	0.115
Mean concave points	0.026	0.088	0.062
Worst radius	13.380	21.135	7.755
Worst area	558.90	1422.29	863.39

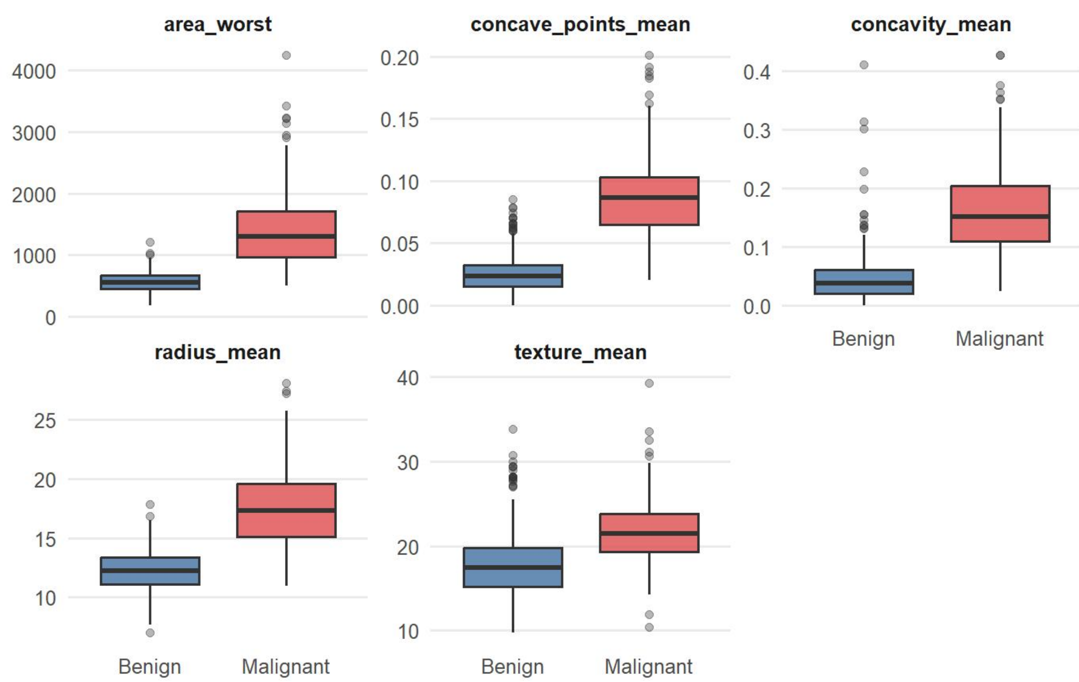

Figure 1. Selected predictors by diagnosis.

Figure 1 shows the same pattern visually. The malignant group is shifted upward for worst area, mean concave points, mean concavity, mean radius, and mean texture. There is still overlap between benign and malignant cases, so no single feature perfectly separates the response. However, the group differences are large enough to suggest that classification models should have meaningful predictive signal.

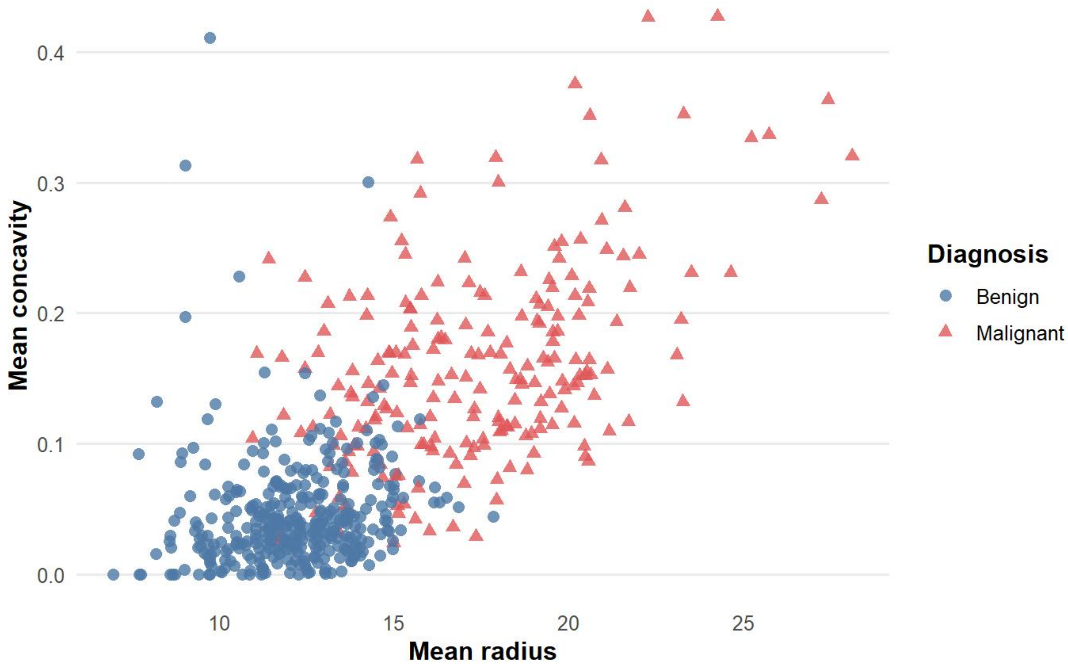

Figure 2. Mean radius and mean concavity by diagnosis.

Figure 2 gives a two-variable view of the class separation. Many malignant cases appear in the upper-right region, where both mean radius and mean concavity are larger. This supports the idea that tumor size and irregularity work together: larger tumors with stronger concavity are more likely to be malignant, although there are still overlapping cases near the center of the plot.

4. Correlated Predictors and Principal Component Analysis

Before fitting the main classification models, it is useful to check how strongly the predictors are related to one another. Many WDBC predictors describe overlapping geometric properties. Radius, perimeter, and area are especially close because they are different measurements of tumor size. The largest correlations and VIF values confirm this structure.

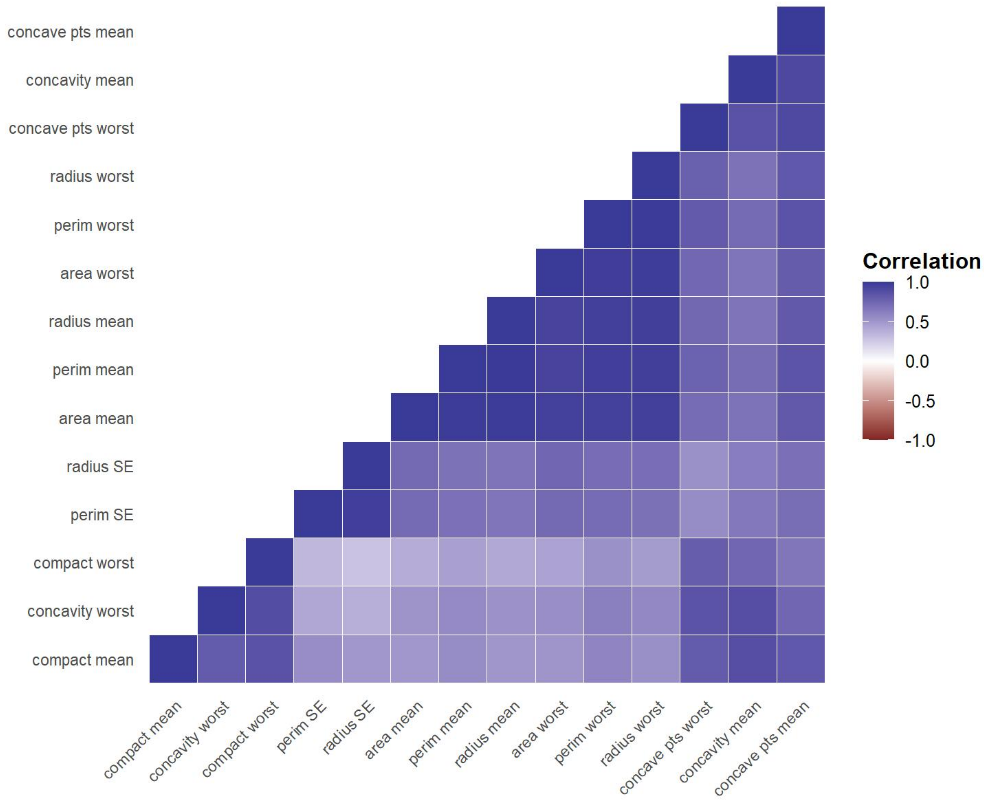

Figure 3. Correlation structure among highly related predictors.

Table 5. Largest absolute correlations.

Variable 1	Variable 2	Correlation
Mean perimeter	Mean radius	0.998
Worst perimeter	Worst radius	0.994
Mean area	Mean radius	0.987
Mean area	Mean perimeter	0.987
Worst area	Worst radius	0.984
Worst area	Worst perimeter	0.978
Perimeter SE	Radius SE	0.973
Mean perimeter	Worst perimeter	0.970

Table 6. Highest VIF values.

Variable	VIF
Mean radius	3806.12
Mean perimeter	3786.40
Worst radius	799.11

Variable	VIF
Worst perimeter	405.02
Mean area	347.88
Worst area	337.22
Radius SE	75.460
Mean concavity	70.770

The highest correlation is between mean perimeter and mean radius (0.998), and the highest VIF values are also for mean radius and mean perimeter. These diagnostics do not mean the variables are useless. Instead, they show that coefficient-level interpretation will be unstable when several closely related measurements are included in the same unpenalized model. This is one reason to include PCA and penalized regression in the analysis.

Principal Component Analysis

Principal component analysis was used as the unsupervised learning component. PCA is appropriate here because all predictors are numeric and many of them measure related aspects of cell size, boundary length, and contour irregularity. All predictors were standardized before PCA. Without standardization, area and perimeter variables would dominate the components largely because of their measurement scale.

Table 7. Proportion of variance explained by the first five principal components.

PC	PVE	Cumulative PVE
PC1	0.443	0.443
PC2	0.190	0.632
PC3	0.094	0.726
PC4	0.066	0.792
PC5	0.055	0.847

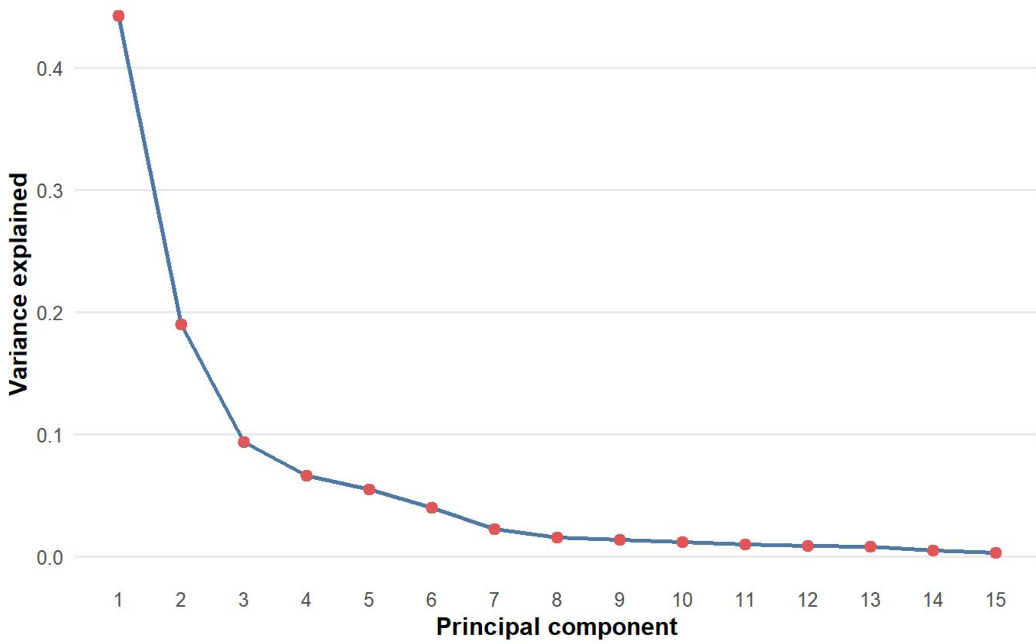

Figure 4. Variance explained by principal component.

The first principal component explains 44.3% of the standardized predictor variance. The second explains 19.0%, so the first two principal components together explain 63.2%. The first five components explain 84.7%. This means that a relatively small number of components captures most of the structure in the 30 predictors.

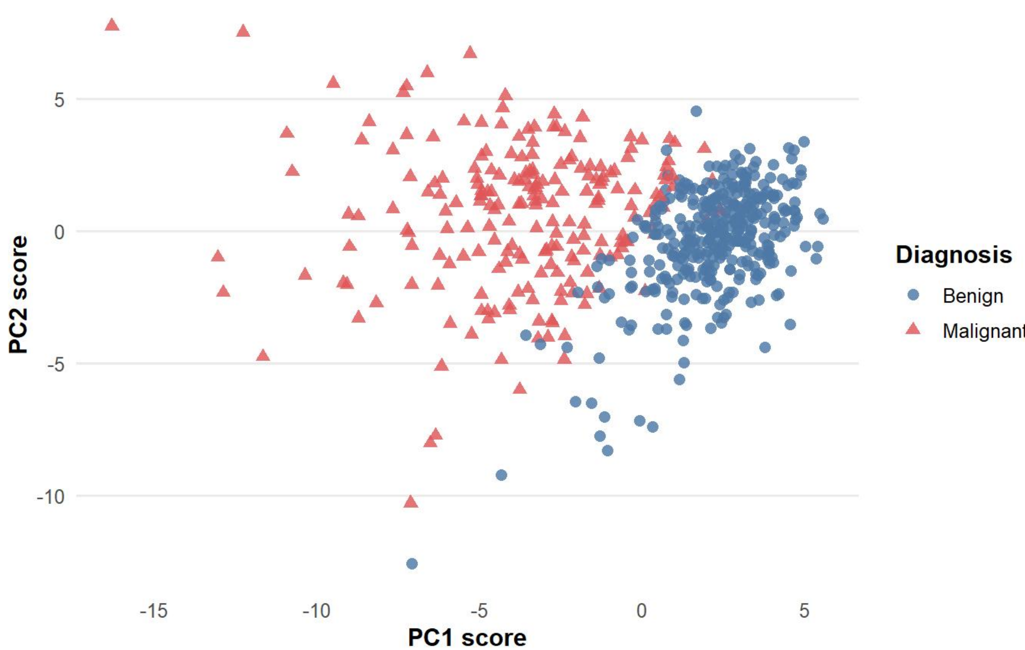

Figure 5. First two PCA scores, colored by diagnosis.

The PCA score plot is fitted without using the diagnosis labels, but coloring the observations by diagnosis shows a clear pattern. Benign and malignant cases are separated mainly along the first principal component. This suggests that the strongest unsupervised direction in the predictors is also closely related to the diagnosis groups.

Table 8. Largest PC1 loadings.

Variable	PC1	PC2	PC3
concave_points_mean	-0.261	0.035	-0.026
concavity_mean	-0.258	-0.060	0.003
concave_points_worst	-0.251	0.008	-0.170
compactness_mean	-0.239	-0.152	-0.074
perimeter_worst	-0.237	0.200	-0.049
concavity_worst	-0.229	-0.098	-0.173
radius_worst	-0.228	0.220	-0.048
perimeter_mean	-0.228	0.215	-0.009

Table 9. Largest PC2 loadings.

Variable	PC1	PC2	PC3
fractal_dimension_mean	-0.064	-0.367	-0.023
fractal_dimension_se	-0.103	-0.280	0.212
fractal_dimension_worst	-0.132	-0.275	-0.233
radius_mean	-0.219	0.234	-0.009
compactness_se	-0.170	-0.233	0.155
area_mean	-0.221	0.231	0.029
radius_worst	-0.228	0.220	-0.048
area_worst	-0.225	0.219	-0.012

PC1 is dominated by concave points, concavity, compactness, perimeter, radius, and area. Because the sign of a principal component is arbitrary, the negative signs are not important by themselves. The important point is that size and irregularity measurements load strongly together. PC2 adds a different direction involving fractal dimension, radius, area, compactness standard error, and smoothness standard error. Together, PC1 and PC2 show that the dataset is not driven by a single raw measurement, but by related groups of size, contour, and texture features.

5. Modeling Strategy

The supervised analysis compares four statistical learning models. The models were chosen to provide both interpretability and predictive performance. A small logistic regression model serves as an interpretable baseline. LDA provides a classical distribution-based classifier. Lasso logistic regression and elastic net logistic regression are included because the predictors are highly correlated and regularization can reduce overfitting while selecting useful features.

Interpretable logistic baseline

The reduced logistic model uses a small set of interpretable predictors: mean radius, mean texture, mean smoothness, mean concavity, and mean concave points. This model is not intended to use all available information. Its purpose is to provide a direct coefficient-based baseline that can be explained through odds ratios.

LDA

Linear discriminant analysis was fit using the full standardized predictor set. LDA estimates class-specific distributions and classifies new observations based on the resulting discriminant rule. It makes assumptions that differ from logistic regression, so it provides a useful comparison point.

Lasso and elastic net logistic regression

Lasso logistic regression was fit with all 30 predictors and a penalty chosen by 10-fold cross-validation. The lasso penalty can set coefficients exactly to zero, which is useful when many predictors are correlated. Elastic net logistic regression extends this idea by mixing lasso and ridge penalties. In this report, elastic net uses alpha = 0.5, giving a balance between sparse selection and grouped shrinkage of correlated variables.

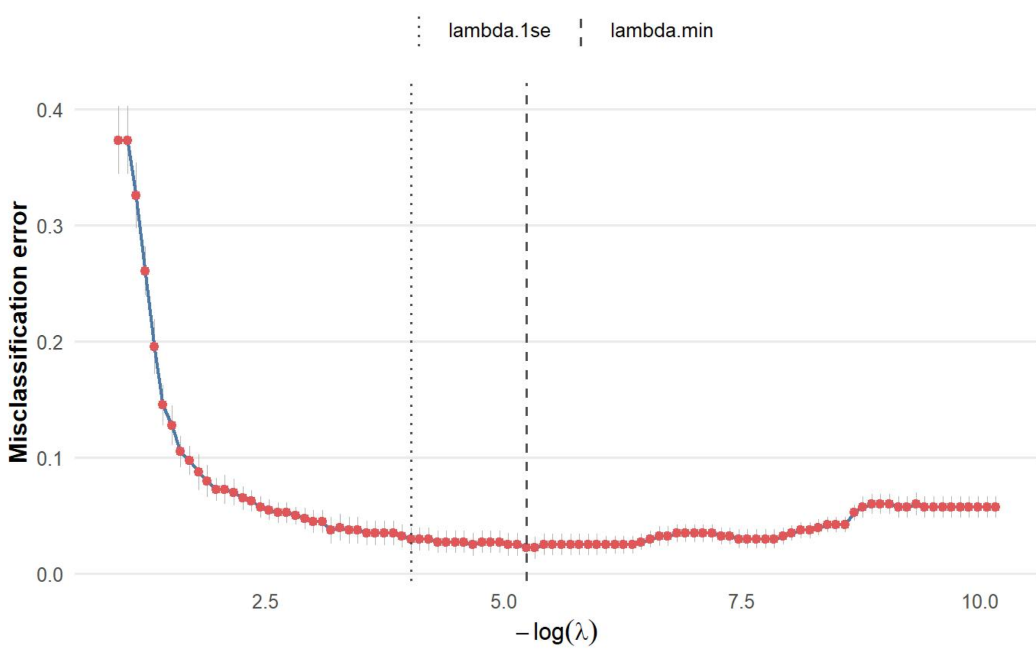

Figure 6. Cross-validation curve for lasso logistic regression.

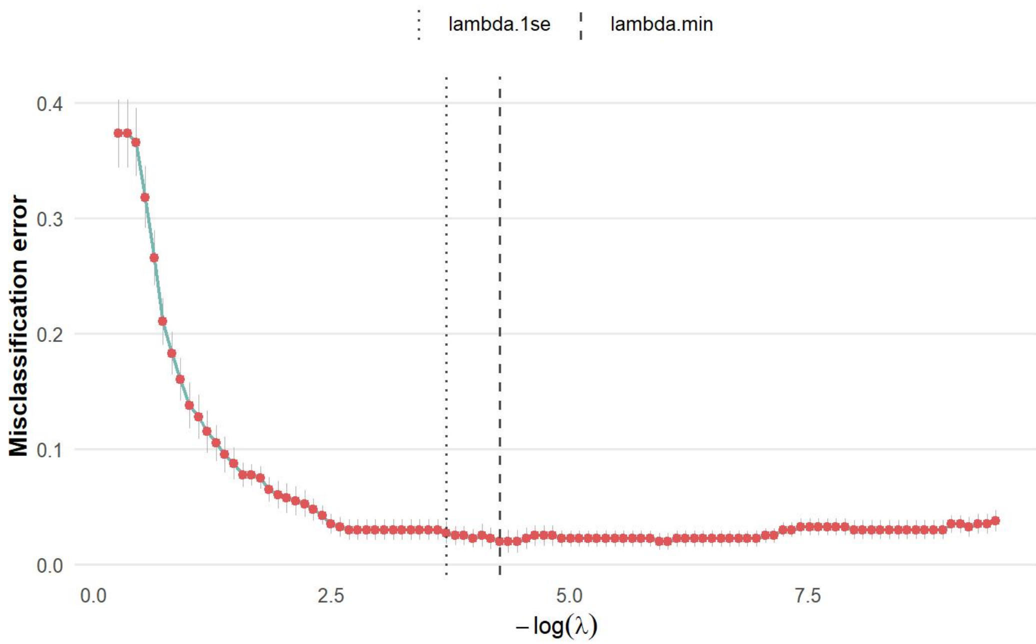

Figure 7. Cross-validation curve for elastic net logistic regression.

For both penalized models, the 1-SE penalty was used for the final comparison. This choice favors a slightly more regularized model that remains close to the cross-validation minimum, rather than selecting the most complex model at the absolute minimum error.

Table 10. Nonzero lasso coefficients at the selected penalty.

Variable	Coefficient
concave_points_worst	17.777
concave_points_mean	13.858
smoothness_worst	11.895
symmetry_worst	3.340
radius_worst	0.489
texture_worst	0.117

Table 11. Largest elastic net coefficients at the selected penalty.

Variable	Coefficient
smoothness_worst	19.453
concave_points_mean	11.549
concave_points_worst	10.181
symmetry_worst	5.289
concavity_mean	1.017
radius_se	0.713
concavity_worst	0.611
radius_worst	0.128
texture_worst	0.101
radius_mean	0.083

The lasso model keeps variables related to worst concave points, mean concave points, worst smoothness, worst symmetry, worst radius, and worst texture. The elastic net keeps a broader set of related size and contour variables. This is expected because elastic net tends to retain groups of correlated predictors more often than pure lasso.

6. Model Evaluation

The models were first evaluated on a held-out test set. Since malignant tumors are the positive class, sensitivity is especially important: a false negative means a malignant tumor is classified as benign. Specificity is also important because it measures how well benign cases are correctly identified. Accuracy and AUC provide additional summaries, but they should be interpreted together with the false-positive and false-negative counts.

Table 12. Holdout test-set model comparison.

Model	Tuning / selection	Accuracy	Test error	Sensitivity	Specificity	AUC
Reduced logistic	Preselected interpretable predictors	0.941	0.059	0.905	0.963	0.991
LDA	No tuning; full standardized predictor set	0.941	0.059	0.873	0.981	0.981
Lasso logistic	10-fold CV, lambda 1-SE	0.959	0.041	0.921	0.981	0.988
Elastic net logistic	10-fold CV, alpha = 0.5, lambda 1-SE	0.971	0.029	0.937	0.991	0.992

Table 13. False-positive and false-negative counts on the test set.

Model	False positives	False negatives
Elastic net logistic	1	4
LDA	2	8
Lasso logistic	2	5
Reduced logistic	4	6

Elastic net logistic regression gives the strongest test-set result, with accuracy 0.971, test error 0.029, sensitivity 0.937, specificity 0.991, and AUC 0.992. It also has the smallest number of false positives and a low number of false negatives. Lasso logistic regression is close behind, with accuracy 0.959 and AUC 0.988. The reduced logistic model and LDA are also strong, but they have weaker sensitivity and more false negatives than elastic net.

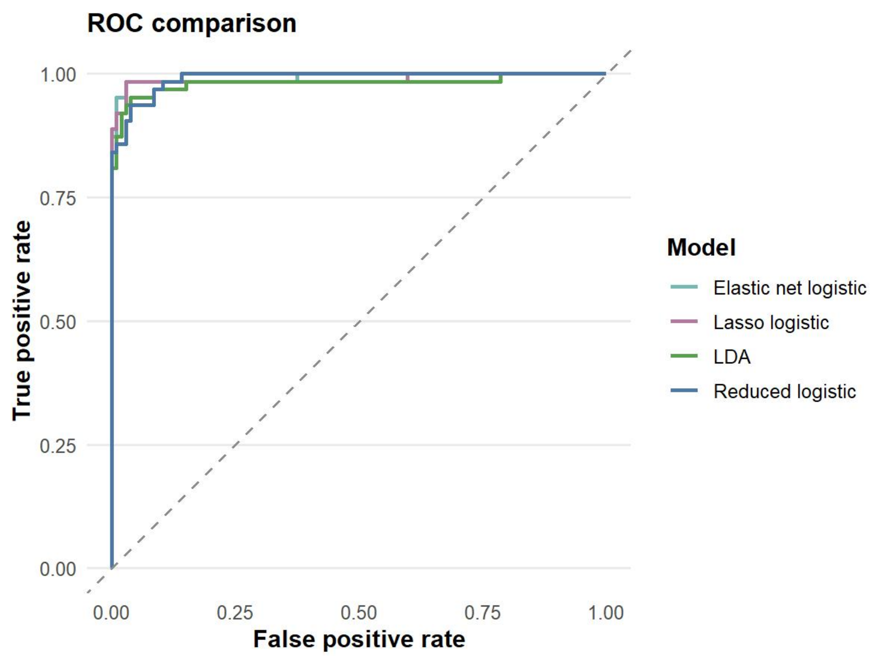

Figure 8. ROC comparison on the holdout test set.

The ROC curves show that all four models separate the two classes well. The differences in AUC are small, so the final model choice should not be based on AUC alone. Elastic net is preferred because it combines the strongest accuracy, the strongest specificity, high sensitivity, and a very strong AUC.

7. Calibration, Thresholds, and Stability

Repeated cross-validation

A single train/test split can be sensitive to random variation. To check whether the model comparison is stable, repeated cross-validation was used as a second evaluation approach. The repeated cross-validation results agree with the holdout comparison.

Table 14. Repeated cross-validation summary.

Model	Accuracy mean	Accuracy SD	Sensitivity mean	Specificity mean	AUC mean	AUC SD
Elastic net logistic	0.972	0.023	0.931	0.996	0.993	0.012
Lasso logistic	0.969	0.024	0.930	0.992	0.993	0.012
LDA	0.957	0.027	0.894	0.994	0.991	0.015
Reduced logistic	0.942	0.038	0.908	0.962	0.985	0.013

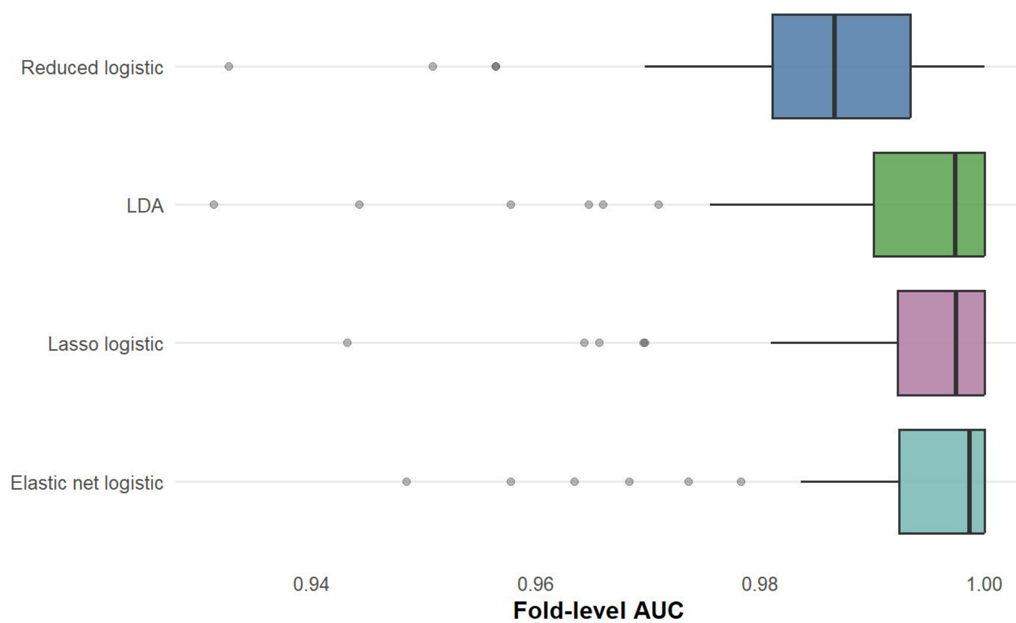

Figure 9. AUC distribution across repeated cross-validation folds.

Elastic net has mean accuracy 0.972 and mean AUC 0.993. Lasso is nearly tied on AUC and only slightly lower in mean accuracy. This suggests that the advantage of elastic net is not only a product of one lucky test split, although the difference between elastic net and lasso should still be described as modest rather than dramatic.

Probability calibration

Classification accuracy measures the final class label, but probability calibration checks whether predicted probabilities behave like probabilities. The Brier score is a useful calibration-related metric, where smaller values indicate better average squared error between predicted probabilities and the observed 0/1 outcome.

Table 15. Brier scores for predicted malignant probabilities.

Model	Brier score
Elastic net logistic	0.029
Lasso logistic	0.030
Reduced logistic	0.037
LDA	0.039

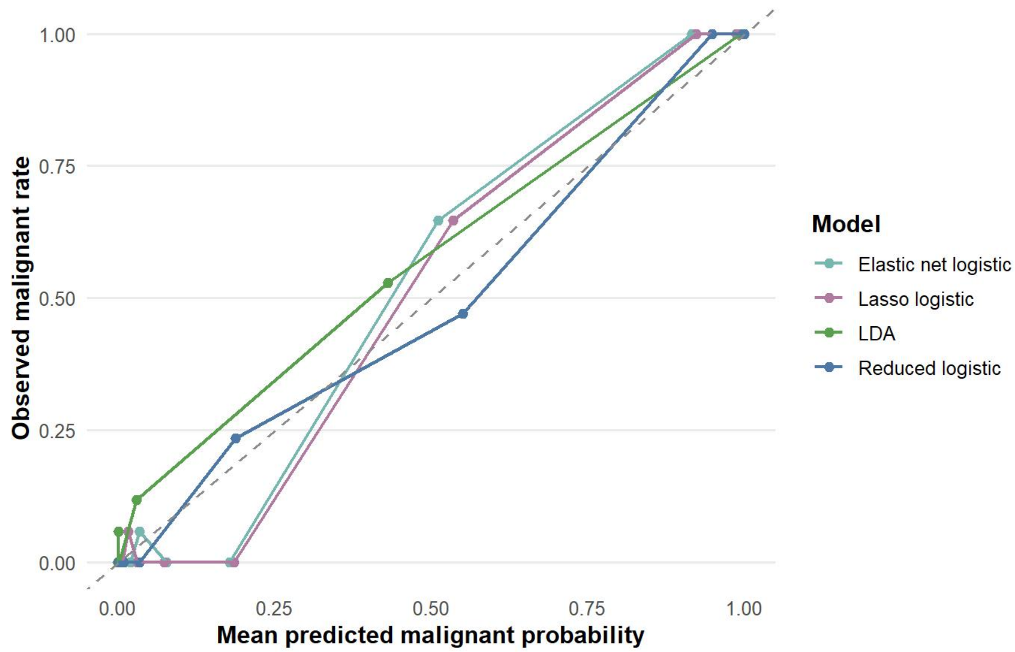

Figure 10. Calibration of predicted malignant probabilities.

Elastic net has the lowest Brier score (0.0294), and lasso is very close (0.0298). The calibration plot shows that the predicted probabilities are generally well aligned with observed outcomes, although the dataset is small enough that calibration should not be overinterpreted.

Threshold choice

The default classification threshold is 0.5, but the threshold can be changed depending on the goal. In a screening context, a lower threshold may be preferred because it can reduce false negatives at the cost of additional false positives. The threshold analysis compares lasso and elastic net at thresholds 0.4, 0.5, and 0.6.

Table 16. Threshold analysis for lasso and elastic net logistic regression.

Model	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1	False negatives	False positives
Lasso logistic	0.400	0.976	0.984	0.972	0.954	0.969	1	3
Lasso logistic	0.500	0.959	0.921	0.981	0.967	0.943	5	2
Lasso logistic	0.600	0.959	0.889	1.000	1.000	0.941	7	0
Elastic net logistic	0.400	0.971	0.968	0.972	0.953	0.961	2	3
Elastic net logistic	0.500	0.971	0.937	0.991	0.983	0.959	4	1
Elastic net logistic	0.600	0.947	0.857	1.000	1.000	0.923	9	0

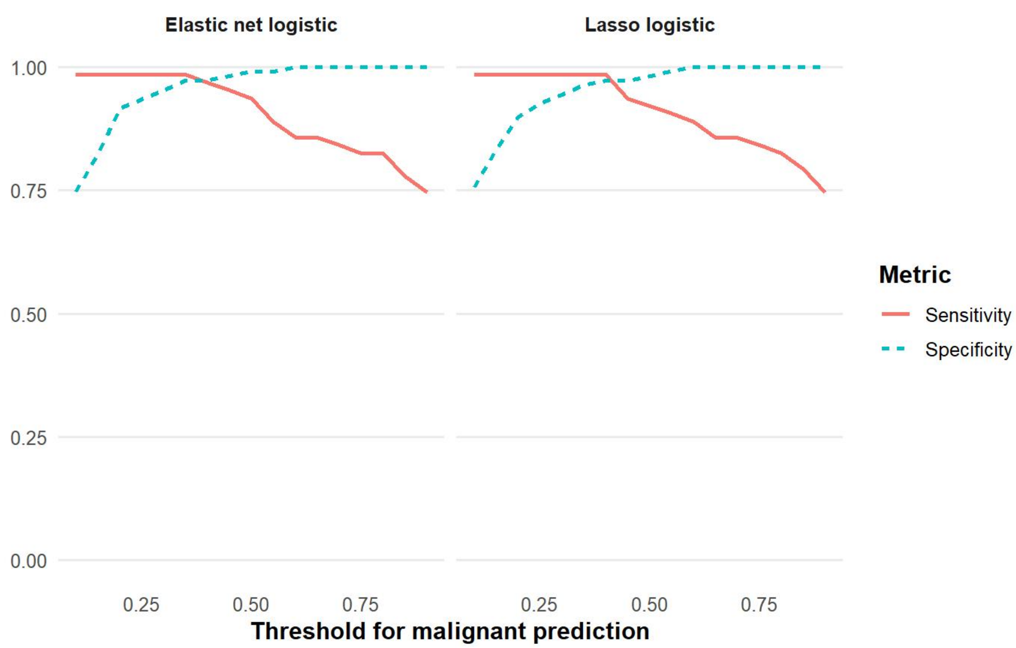

Figure 11. Sensitivity and specificity across thresholds.

For lasso, lowering the threshold from 0.5 to 0.4 reduces false negatives from five to one, while false positives increase from two to three. For elastic net, lowering the threshold from 0.5 to 0.4 reduces false negatives from four to two, while false positives increase from one to three. This tradeoff is important because the cost of a false negative is likely higher in a medical screening setting.

Variable-selection stability

Variable selection was checked with 100 bootstrap refits. Each refit recorded whether a variable was selected by the penalized model. This does not prove that selected variables have causal effects, but it helps distinguish variables that appear consistently from variables that only appear under one sample split.

Table 17. Selected bootstrap stability results.

Model	Variable	Times selected	Selection frequency
Elastic net logistic	area_mean	100	1.000
Elastic net logistic	area_worst	100	1.000
Elastic net logistic	concave_points_mean	100	1.000
Elastic net logistic	concave_points_worst	100	1.000
Elastic net logistic	perimeter_mean	100	1.000
Elastic net logistic	perimeter_worst	100	1.000
Lasso logistic	concave_points_worst	100	1.000
Lasso logistic	texture_worst	100	1.000
Lasso logistic	radius_worst	98	0.980
Lasso logistic	smoothness_worst	93	0.930
Lasso logistic	symmetry_worst	84	0.840
Lasso logistic	concave_points_mean	82	0.820

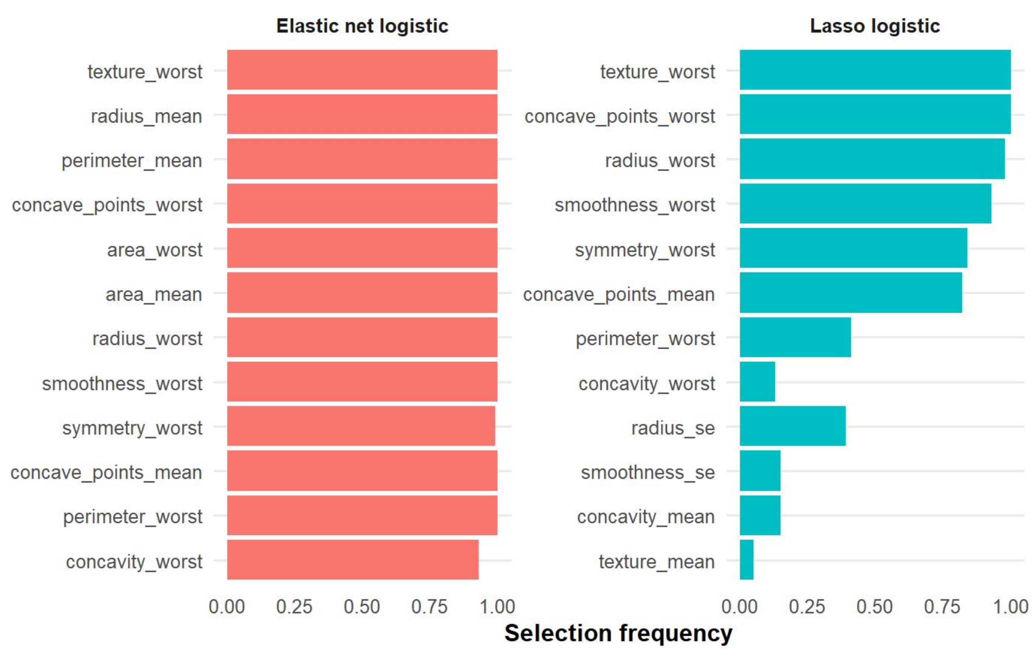

Figure 12. Selection frequencies from bootstrap refits.

Several elastic net variables have selection frequency 1.00, including mean area, worst area, mean concave points, worst concave points, mean perimeter, worst perimeter, mean radius, and worst radius. The lasso model is more selective but repeatedly keeps worst concave points and worst texture. Overall, the stability results match the EDA and PCA interpretation: size, contour irregularity, and texture are the strongest recurring signal.

8. Interpretation

The reduced logistic model is the easiest model to interpret directly because it is fit on a small number of original-scale predictors. Two coefficients are especially useful for explaining the direction of the fitted relationship.

Table 18. Interpretable odds ratios from the reduced logistic model.

Predictor	Estimate	Std. error	p-value	Odds ratio
Mean radius	0.893	0.221	<0.001	2.441
Mean texture	0.318	0.070	<0.001	1.375

Holding the other included predictors fixed, a one-unit increase in mean radius is associated with multiplying the estimated odds of malignancy by about 2.44. A one-unit increase in mean texture is associated with multiplying the estimated odds of malignancy by about 1.37. These interpretations are conditional associations, not causal effects.

The coefficient interpretation should be read carefully because many WDBC predictors are strongly correlated. When radius, area, perimeter, concavity, and concave points are included together, each coefficient represents the effect of changing one predictor while holding the related predictors fixed. That kind of conditional interpretation can be unstable when predictors measure overlapping geometric properties. This is why the reduced logistic model is best used as an interpretation baseline, while elastic net is preferred for prediction.

9. Technical Note: Standardization and Penalized Regression

Standardization is a practical part of this analysis, not just a cosmetic change. The WDBC predictors are measured on very different scales: area-related variables can be in the hundreds or thousands, while smoothness and fractal-dimension variables are small decimals. For methods based on distances, variance, or penalties, these scale differences affect the fitted model unless the predictors are standardized first.

For one predictor, standardization can be written as:

$$z_j = (x_j - \bar{x}_j) / s_j$$

If the fitted model uses standardized predictors, the linear predictor can be written as:

$$\eta = \alpha_0 + \sum_{j=1}^{p} \alpha_j z_j$$

The same fitted linear predictor can also be written on the original predictor scale:

$$\eta = \beta_0 + \sum_{j=1}^{p} \beta_j x_j$$

The relationship between the two sets of coefficients is:

$$\beta_j = \alpha_j / s_j, \quad \beta_0 = \alpha_0 - \sum_{j=1}^{p} (\alpha_j \bar{x}_j / s_j)$$

This conversion is useful for interpretation because it separates two ideas. Standardization changes the coordinate system used during fitting, while the linear predictor can still be translated back to the original units after fitting. For lasso and elastic net, standardization is especially important before fitting because the penalty is applied to the numerical coefficients. Without standardization, a variable measured on a large scale could receive a smaller numerical coefficient and be penalized differently from a variable measured on a small scale.

In this project, PCA, LDA, lasso logistic regression, and elastic net logistic regression use standardized predictors. The reduced logistic model is kept on the original scale because it is mainly used for odds-ratio interpretation. This is why the predictive models and the interpretation baseline are presented separately.

10. Limitations

There are several limits to the analysis. First, the WDBC dataset is clean, well-known, and relatively small. Model performance may be lower in a noisier clinical workflow or in data collected from a different population. Second, even though repeated cross-validation was used, the project still relies on one public dataset. External validation on a separate dataset would be needed before making stronger claims about generalization. Third, the predictors are highly correlated, so individual coefficient estimates and selected variables should not be interpreted as isolated biological effects. Finally, the model is a statistical classifier, not a clinical decision rule. A real diagnostic workflow would need clinical validation, careful threshold selection, and domain-specific review.

11. Conclusion

The Wisconsin Diagnostic Breast Cancer measurements provide strong information for separating malignant and benign tumors. Exploratory analysis shows clear differences in size and contour-irregularity features, and PCA shows that the strongest unsupervised variation is closely related to those same measurements. The supervised models all perform well, but elastic net logistic regression gives the strongest overall performance in this analysis. On the holdout test set, elastic net achieves accuracy 0.971, sensitivity 0.937, specificity 0.991, and AUC 0.992. Repeated cross-validation supports this result with mean accuracy 0.972 and mean AUC 0.993.

The model comparison suggests that regularized logistic regression is well suited to this dataset because it can use the full predictor set while handling strong correlation among related measurements. The reduced logistic model remains useful for communication because its odds ratios provide a direct interpretation of variables such as mean radius and mean texture. Taken together, the results show how exploratory analysis, PCA, regularized classification, calibration, threshold analysis, and stability checks can support one coherent statistical learning report for a binary medical-classification dataset.

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